

Day 8 of training

What is JavaScript?

JavaScript is a programming language used to make webpages interactive.

JavaScript is a programming language that helps websites respond to actions, move things, and interact with users.

Language	What it does	
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HTML	Makes the content	
CSS	Makes it look good	
JavaScript	Makes it work & interact	

Advantages of JavaScript

Fast

→ Runs directly in the browser — no need to wait!

Easy to Learn

→ Looks like English, perfect for beginners.

Makes Webpages Interactive

→ Pop-ups, sliders, buttons, forms — all possible!

Works Everywhere

→ Runs on all modern browsers (Chrome, Firefox, Safari, etc.)

No Extra Software Needed

→ Just open your browser and write JavaScript.

Used for Frontend and Backend

→ Can build both website designs and server logic (using Node.js).

Disadvantages of JavaScript

Browser Dependent

→ Sometimes behaves differently in different browsers.

Security Issues

→ Can be used to write harmful code if not handled properly.

Can Be Slow for Big Apps

→ Not as fast as some other languages for complex programs.

No File Access (Directly)

→ Can't directly read/write files on your computer (for safety).

Easily Misused

→ Beginners might write messy or buggy code.

Applications of JavaScript

JavaScript is used to make websites alive and interactive. But that's not all — it's used in many areas!

1. Web Development

Add buttons, sliders, popups, animations, etc.

Example: Click a button to open a form

2. E-commerce Websites

Add to cart, price calculators, discount updates

Example: Amazon cart updates without reloading

4. Mobile App Development

Used with frameworks like React Native

Build apps for Android and iOS using JS

Comments

there are two types of comments in JavaScript

1) single line comments (using `->//`)

2) multiple line comments (using `-> /* */`)

IDENTIFIERS IN JS

identifiers in js is same as other language like c,c++

there are some rules for identifiers

--> Must start with a letter (A-Z, a-z) or underscore (`_`)

--> Can contain letters, digits (0-9), and underscores

--> No special characters allowed (like `@`, `$`, `#`, `!`)

--> Cannot be a keyword

What is a Variable?

A variable is like a container or a box that stores some information which can change.

What are Keywords in JavaScript?

Keywords are special reserved words in JavaScript that have special meaning.

You cannot use them as variable names, because JavaScript already uses them for its own purpose.

Keyword	Meaning
let	Declare a block-scoped variable
const	Declare a constant (unchangeable) variable
var	Declare a variable (old method)